JWST/NIRCam F150W Surface Brightness Fluctuation Distances from GO-3055: A TRGB-Anchored Calibration and Internal Accuracy Test

I. A. Bukanov
Lomonosov Moscow State University, Faculty of Space Research,
Moscow, Russia

Co-author list to be agreed

Draft of 1 August 2026

Abstract

We measure JWST/NIRCam F150W surface brightness fluctuation (SBF) distances for 14 early-type galaxies in program GO-3055 and calibrate them directly with the individual JWST tip of the red giant branch (TRGB) distances from Chazov et al. (2026). The primary experiment is fixed in advance: all 14 galaxies, individual Paper IV thin-slice distances, no manual rejection, and one color-independent F150W zero point. A linear color term is tested before the distance analysis, but is not adopted because it does not improve either the information criterion or cross-validation. The resulting calibration is $\overline{M}_{150,0} = -3.172$ mag with fitted intrinsic scatter 0.084 mag. Its leave-one-out (LOO) distance-modulus bias is -0.0004 mag and its RMS is 0.0955 mag, corresponding to 4.4 per cent in distance. The median predicted internal uncertainty is 0.0902 mag (4.2 per cent); including the common TRGB absolute-scale uncertainty gives 0.1018 mag (4.8 per cent). The normalized LOO residuals have RMS 1.08; nine of 14 objects are within 1σ and all 14 are within 2σ. For comparison, a linear color law gives a worse LOO RMS of 0.0992 mag, is disfavored by ΔAICc = 2.58, and has a bootstrap slope interval that includes zero. The data therefore do not require a color dependence, and all primary distances and uncertainties below use the constant zero point. Paper IV thick-slice distances and the Paper III cluster-mean prescription give consistent accuracy. The result is an internal cross-validation of the calibrators, not an independent external validation.

Keywords: distance scale; galaxies; surface brightness fluctuations; JWST; NIRCam; TRGB.

1 Introduction

Surface brightness fluctuations arise from Poisson variations in the number and luminosity of unresolved stars within each resolution element [1]. Their apparent amplitude decreases with distance, so an absolute SBF magnitude calibrated for the stellar population provides a distance indicator. In the near-infrared, evolved stars produce a strong SBF signal, but the absolute amplitude may depend on age and metallicity. Color is therefore commonly used as a population correction; it is an optional improvement to the distance estimator, not the definition of the method.

The TRGB–SBF project is constructing a Population-II distance ladder that transfers a geometrically calibrated TRGB zero point to SBF. Jensen et al. [Paper III; 13] used JWST TRGB distances to recalibrate the existing HST/WFC3-IR F110W SBF scale. With mean Virgo and Fornax distances, they obtained

$$\overline{M}_{110} = (1.86 \pm 0.16)[(g - z)_{\rm PS} - 1.30] - (2.760 \pm 0.016).$$

Here the SBF signal is measured from the JWST GO-3055 F150W images. The F090W–F150W color is measured independently only to test whether a population correction is required; it is not an input to the F150W power-spectrum measurement. The anchors are the revised JWST TRGB distances of Chazov et al. [Paper IV; 14], tied to the NGC 4258 maser distance. Our main questions are deliberately ordered as follows: (1) is a color term required; (2) if not, how accurately does a single F150W SBF zero point recover the TRGB distances; and (3) is the propagated uncertainty consistent with the observed residuals?

2 Data and distance anchors

2.1 GO-3055 sample

The sample contains 14 galaxies: three Fornax members, nine galaxies in or near the Virgo region, and two nearby group galaxies. Every target has NIRCam F090W and F150W imaging. The signal and color are measured in the same two circular annuli, 8.2–16.4 and 16.4–32.8 arcsec, following the geometry of Paper III. At the individual Paper IV distances, the outer 32.8-arcsec radius spans 1.65–3.17 kpc. Consequently, radial differences cannot be interpreted as measurements at a common physical radius. Jensen et al. also used an auxiliary elliptical-annulus check for elongated galaxies; that secondary test is not reproduced here, and the limitation is kept explicit.

The primary anchors are the individual Paper IV thin-slice TRGB moduli. This choice was fixed before inspection of the present SBF residuals and is kept to avoid selecting an anchor variant after seeing which gives less scatter. Paper IV prefers its thick-slice estimator because it uses a broader and more robust stellar sample on the blue side of the TRGB color break. We therefore treat thick-slice distances as a mandatory sensitivity experiment rather than silently ignoring that preference.

Two Jensen comparisons are kept separate throughout. Paper III is the HST F110W calibration of 14 TRGB–SBF targets and is used for the color-law and cluster-mean comparisons. A legacy Jensen et al. (2015) F160W reconstruction exists for only five common galaxies and is reported only as a separately labelled distance column; it is never merged with the Paper III F110W sample.

2.2 Foreground extinction

Paper IV tabulates A_{090}. Using its coefficients, we adopt

$$E(B-V) = A_{090}/1.4156, \tag{1}$$
$$A_{150} = 0.6021E(B-V), \tag{2}$$
$$C_0 = C_{\rm obs} - (A_{090} - A_{150}), \tag{3}$$
$$\overline{m}_{150,0} = \overline{m}_{150,\rm obs} - A_{150}. \tag{4}$$

The corrections make the colors bluer by 0.008–0.029 mag and the F150W SBF magnitudes brighter by 0.006–0.022 mag. They do not materially change the distance RMS, but they are required for an unbiased zero point and covariance matrix.

No separate internal-extinction correction is applied. The central 8.2 arcsec are excluded, obvious non-elliptical features enter the object mask, and the sample contains no retained strong dust lane. As a quantitative sensitivity bound, an unrecognized screen with $A_V^{\rm int} \leq 0.05$ mag would give $A_{150}^{\rm int} \lesssim 0.01$ mag for a standard near-infrared extinction curve, well below the observed 0.0955-mag LOO scatter. This is a bound, not a measurement of internal dust.

All targets have $z < 0.007$. For a deliberately broad local spectral-slope range $|1+\alpha| \leq 3$, where $f_\nu \propto \nu^\alpha$, the ordinary same-filter correction satisfies

$$|K_{150}| = 2.5|1+\alpha| \log_{10}(1+z) < 0.023 \text{ mag}. \tag{5}$$

It is not applied object by object because a consistent SBF spectral-energy model is unavailable and the bound is subdominant at the present precision. Likewise, cosmological surface-brightness dimming is not added as a separate SBF term: the common surface-brightness factor cancels in the variance divided by mean-galaxy-light normalization. Even the full Tolman term is bounded by $10\log_{10}(1+z) < 0.030$ mag over the sample. These two bounds delimit, rather than conceal, the small-redshift approximation.

3 SBF measurement

3.1 Sky, galaxy model, and masks

The input images are the public calibrated NIRCam i2d mosaics in MJy sr^{-1}. A scalar sky is estimated outside the science annuli from independent background regions and subtracted. Invalid pixels, detector gaps, foreground objects, and compact sources form the primary mask. Initial compact sources require at least nine connected pixels above a local 3.5σ threshold; after galaxy subtraction, a second catalogue uses four connected pixels above 2.5σ, deblending, and a three-pixel morphological dilation. The mask is deliberately aggressive toward compact positive residuals while retaining diffuse galaxy structure.

The centre is initialized from the image and checked against the WCS target position. Ellipticity and position angle remain free while isophotes are fitted from a starting semimajor axis of 50 pixels to at most 2000 pixels. Intensities and fitted geometry are linearly interpolated between consecutive isophotes; there is no piecewise constant filling between isophotal lines. A Sersic image is used only to fill detector gaps while solving the isophotes. The final model at every real science pixel comes from the measured isophotal profile. Filled gap pixels never enter a power spectrum.

After subtraction, residual values in the science region are limited symmetrically to the robust median plus or minus 3.5 robust standard deviations. Pixels are not rejected. This operation is therefore *winsorization*, not sigma-clipping; sigma-clipped statistics are used only to estimate its centre and scale. The raw and winsorized residual products are both retained for diagnostics.

3.2 Normalized residual and expectation spectrum

For a smooth model M_i, sky-subtracted data I_i, and binary usable-pixel window W_i, the field entering the Fourier analysis is

$$R_i = W_i \frac{I_i - M_i}{\sqrt{M_i}}. \tag{6}$$

The division by $\sqrt{M_i}$ converts the Poisson stellar variance, which is proportional to local galaxy intensity, into an approximately stationary SBF amplitude. It does not measure a difference between the brightest and faintest pixels. SBF is the variance of the unresolved stellar luminosity function per unit mean light, recovered statistically from all usable pixels.

For each annulus we subtract the mean of R, calculate the two-dimensional FFT, and use

$$P(\boldsymbol{k}) = \frac{|\mathcal{F}[R](\boldsymbol{k})|^2}{N_{\rm use}}. \tag{7}$$

The spectrum is azimuthally averaged into 80 radial bins; the uncertainty of each bin is the standard error of its contributing Fourier modes. The expected SBF spectrum is not simply $|\widetilde{\rm PSF}|^2$. For each of five time- and position-matched, unit-normalized STPSF models, 64 independent unit white-noise fields are convolved with the PSF, multiplied by the exact annular window and source mask, mean-subtracted, Fourier transformed, and averaged:

$$E_j(k) = \left\langle \frac{|\mathcal{F}\{W[n * {\rm PSF}_j]\}|^2}{N_{\rm use}} \right\rangle_{64}. \tag{8}$$

Thus $E(k)$ includes PSF blurring, the finite annulus, holes in the source mask, and mode mixing by the window.

Table 1: Extinction-corrected GO-3055 measurements. The penultimate column is the color-independent LOO SBF-minus-TRGB residual. HQ IV reproduces the high-quality flag in Table 2 of Paper IV. The quoted $\sigma_{\overline{M}}$ contains individual errors but excludes common zero points. The annular half-difference is shown as a physical/quality diagnostic and is not included as random noise.

Galaxy	Environment	$C_{090-150,0}$ (mag)	$\overline{m}_{150,0}$ (mag)	$\overline{M}_{150,0}$ (mag)	$\sigma_{\overline{M}}$ (mag)	$\lvert\Delta\overline{m}\rvert_{\rm ann}$ (mag)	$\mu_{\rm TRGB}$ (mag)	$\Delta\mu_{\rm LOO,const}$ (mag)	HQ IV
NGC 1380	Fornax	0.510	28.111	-3.297	0.029	0.046	31.408	-0.135	+
NGC 1399	Fornax	0.570	28.180	-3.318	0.034	0.019	31.498	-0.157	−
NGC 1404	Fornax	0.559	28.206	-3.160	0.028	0.026	31.366	+0.014	+
NGC 1549	Other	0.533	27.986	-3.245	0.029	0.046	31.231	-0.079	−
NGC 3379	Other	0.557	27.042	-3.036	0.030	0.168	30.078	+0.147	+
NGC 4374	Virgo	0.566	27.965	-3.147	0.031	0.043	31.112	+0.027	+
NGC 4406	Virgo	0.616	27.863	-3.233	0.030	0.062	31.096	-0.065	+
NGC 4472	Virgo	0.625	27.763	-3.253	0.035	0.039	31.016	-0.087	−
NGC 4486	Virgo	0.652	27.878	-3.168	0.054	0.069	31.046	+0.005	−
NGC 4552	Virgo	0.569	27.876	-3.047	0.030	0.086	30.923	+0.134	+
NGC 4621	Virgo	0.565	27.717	-3.104	0.028	0.072	30.821	+0.074	+
NGC 4636	Virgo	0.604	27.926	-3.144	0.031	0.022	31.070	+0.030	+
NGC 4649	Virgo	0.623	27.961	-3.048	0.030	0.119	31.009	+0.133	−
NGC 4697	Virgo	0.512	27.109	-3.215	0.032	0.114	30.324	-0.046	+

3.3 Power-spectrum fit and fluctuation magnitude

For every PSF realization we fit

$$P(k) = P_{0,j}E_j(k) + P_{1,j} \tag{9}$$

by weighted least squares, with weights equal to the inverse squared bin errors. The covariance is $(X^{\top}WX)^{-1}$ multiplied by $\max(\chi^2_\nu, 1)$. The adopted P_0 and P_1 are the medians over the five PSFs. The primary normalized-frequency interval is $0.04 \le k \le 0.25$, with an additional lower bound of ten Fourier modes across the shorter crop axis. Fits beginning at $k_{\rm min} = 0.01$ and 0.03, with the same $k_{\rm max}$, quantify low-frequency sensitivity but do not replace the primary fit.

The PSF-shaped power contains both stellar SBF and unresolved compact sources:

$$P_{\rm SBF} = P_0 - P_r. \tag{10}$$

Here P_r is specifically the contribution from unmasked globular clusters and background galaxies fainter than the catalogue limit. In each annulus the observed source luminosity function is modelled as a Gaussian globular-cluster luminosity function plus a power-law background-galaxy term. Its faint side is integrated with flux-squared weighting,

$$P_r = \langle M^{-1}\rangle \int_{m_{\rm cut}}^{m_{\rm cut}+12} [\phi_{\rm GC}(m) + \phi_{\rm bg}(m)] f^2(m)\, dm, \tag{11}$$

where $f(m)$ is the flux in image units and $\langle M^{-1}\rangle$ restores the normalization of Eq. 6. The adopted GCLF width is 1.2 mag; widths of 1.35 mag for NGC 1399 and NGC 4472 are tested separately.

Because Eq. 6 already divides the variance by mean galaxy light, $P_{\rm SBF}$ has units of fluctuation flux and is not divided by M a second time. With pixel solid angle $\Omega_{\rm pix}$,

$$Z_{\rm AB,pix} = -2.5\log_{10}\left(\frac{10^6\Omega_{\rm pix}}{3631}\right), \tag{12}$$

$$\overline{m}_{150} = -2.5\log_{10}(P_0 - P_r) + Z_{\rm AB,pix}. \tag{13}$$

The inner and outer magnitudes are finally combined with inverse-variance weights. Their half-difference is retained as a radial diagnostic, not inserted as random measurement noise.

All current annular fits give $P_1 < 0$. Corner regions of the original F150W i2d frames have a median logarithmic noise-spectrum slope of -1.45 over the fitted frequency range, so the data are not white. The fitted constant absorbs the local intercept of this misspecified noise spectrum and may be negative; it is not a claim of negative physical noise power.

3.4 Matched color

F090W is registered and PSF-matched to F150W. Color is measured in the same usable pixels and rings as the SBF signal. The two annular colors may differ because the stellar population changes with radius. Their half-difference is therefore retained as a radial diagnostic, not assigned as a statistical color error. Color does not enter the primary power-spectrum measurement or the adopted distance estimator; it is used only in the model-comparison test of a possible population correction.

4 Uncertainty model

The uncertainty bookkeeping separates errors that vary from galaxy to galaxy, finite-calibration uncertainty, intrinsic population scatter, and common absolute zero points. This separation prevents a common systematic from being incorrectly used to down-weight an individual point.

4.1 Calibration-point errors

Let r denote the inner or outer annulus. The non-PSF power-spectrum error is defined conservatively as

$$\sigma_{k,r} = 1.4826\,\mathrm{MAD}\{\overline{m}_r(k_{\min})\}, \tag{14}$$

$$\sigma_{\mathrm{WLS},r} = \frac{1.085736\,\sigma_{P_0,\mathrm{WLS}}}{P_0 - P_r}, \tag{15}$$

$$\sigma_{\mathrm{PS},r} = \max(\sigma_{k,r}, \sigma_{\mathrm{WLS},r}). \tag{16}$$

where the MAD uses $k_{\min} = 0.01, 0.03$, and 0.04 and the second term is the linear propagation of the WLS covariance. It is a maximum, not a quadrature sum, because both terms diagnose uncertainty in the same fitted amplitude. The factor $1.085736 = 2.5/\ln 10$ converts fractional power error to magnitude.

The five PSFs are then kept separate. Their robust scatter is

$$\sigma_{\mathrm{PSF},r} = 1.4826\,\mathrm{MAD}\{\overline{m}_{r,j}\}. \tag{17}$$

The analysis notebook reconstructs Eq. 16 from the saved $\sigma_{P_0,\mathrm{WLS}}$ before adding this term; hence PSF scatter is not counted both inside and outside the fit error. With the same normalized inverse-variance annular weights w_r used for the central magnitude,

$$\sigma^2_{\mathrm{PS}} = \sum_r w_r^2 \sigma^2_{\mathrm{PS},r}, \tag{18}$$

$$\sigma_{\mathrm{PSF}} = \sum_r w_r \sigma_{\mathrm{PSF},r}. \tag{19}$$

The PSF terms are combined linearly because a single PSF mismatch moves both annuli coherently.

The scalar-sky systematic s is the scatter among independent background estimates in image units. Linearizing Eq. 13 gives

$$\sigma_{\mathrm{sky},r} = 1.085736\frac{s}{\langle M\rangle_r}, \qquad \sigma_{\mathrm{sky}} = \sum_r w_r \sigma_{\mathrm{sky},r}. \tag{20}$$

Again the annuli are correlated because the same scalar sky is subtracted from both. If $q_r = P_r/P_0$, assigning the stated 25 per cent fractional uncertainty to P_r gives

$$\sigma_{P_r,r} = 1.085736\,(0.25)\frac{q_r}{1-q_r}, \qquad \sigma_{P_r} = \sum_r w_r \sigma_{P_r,r}. \tag{21}$$

This term is small because $P_r/P_0 < 0.0067$ for every retained annulus; the 0–$2P_r$ sensitivity experiment tests the stronger possibility that the working luminosity-function model itself is wrong.

The internal apparent-SBF error is therefore

$$\sigma^2_{\overline{m},\mathrm{int}} = \sigma^2_{\mathrm{PS}} + \sigma^2_{\mathrm{PSF}} + \sigma^2_{\mathrm{sky}} + \sigma^2_{P_r}. \tag{22}$$

The radial half-difference does not appear in Eq. 22: it can be a real population gradient and the two rings share their sky, model, mask, and PSF.

For foreground reddening we assign

$$\sigma^2_{A_{090}} = (0.16A_{090})^2 + 0.005^2, \tag{23}$$

$$\sigma_{E(B-V)} = \sigma_{A_{090}}/1.4156, \tag{24}$$

$$\sigma_{E(090-150)} = (1.4156 - 0.6021)\sigma_{E(B-V)}. \tag{25}$$

The individual Paper IV modulus error already contains its A_{090} term. We remove that contribution, $\sigma^2_{\mu,\mathrm{no\ red}} = \sigma^2_{\mu,\mathrm{IV}} - \sigma^2_{A_{090}}$, and then restore the single correct shared-reddening combination. The error of an absolute calibration point is

$$\sigma^2_{\overline{M},i} = \sigma^2_{\overline{m},\mathrm{int}} + \sigma^2_{\mu,\mathrm{no\ red}} + \sigma^2_{E(090-150)}. \tag{26}$$

The same reddening moves corrected color and absolute SBF in opposite directions, giving

$$\mathrm{Cov}(C_0, \overline{M}_{150,0}) = -\sigma^2_{E(090-150)}. \tag{27}$$

The photometric color error in each ring is the standard error of the usable pixel-color distribution after WCS registration and PSF matching. The two ring errors are combined as $\sigma^2_{C,\mathrm{phot}} = \sum_r w_r^2\sigma^2_{C,r}$. The horizontal error used in the diagnostic linear regression is $\sigma^2_C = \sigma^2_{C,\mathrm{phot}} + \sigma^2_{E(090-150)}$. This is a photometric error; the observed inner–outer color difference remains a separate radial-gradient diagnostic. The standard error does not claim to include every image-wide F090W/F150W calibration or PSF-matching systematic. Adding a deliberately larger 0.01-mag independent color floor leaves the linear LOO RMS unchanged at the reported precision, so this omission cannot drive the present conclusion that the color term is not required.

4.2 Zero-point calibration and predictive distance errors

The adopted calibration is color independent,

$$\overline{M}_{150,0} = a, \qquad \mu_{\mathrm{SBF}} = \overline{m}_{150,0} - a. \tag{28}$$

Consequently, neither the measured color nor its uncertainty enters a primary distance. A color law $f(C)$ is fitted only as a model-comparison diagnostic.

For a color law $f(C)$, the likelihood uses the effective variance

$$V_i = \sigma^2_{\overline{M},i} + [f'(C_i)\sigma_{C,i}]^2 - 2f'(C_i)\mathrm{Cov}(C_i, \overline{M}_i) + \sigma^2_{\mathrm{int}}. \tag{29}$$

and maximizes the Gaussian likelihood including $\ln V_i$. For the adopted constant relation, $f'(C) = 0$, so all color and color-covariance terms vanish. The intrinsic term σ_{int} is therefore fitted, not chosen to force reduced $\chi^2 = 1$. Fits with no horizontal error, an additional 0.01-mag object-to-object floor, and the historical half-ring proxy are reported as sensitivity tests.

For a new target under the adopted constant calibration, the target-measurement component is simply

$$\sigma^2_{\mathrm{target,const}} = \sigma^2_{\overline{m},\mathrm{int}} + \sigma^2_{A_{150}}. \tag{30}$$

For completeness, the corresponding expression used only in the diagnostic color-law fit is

$$\sigma^2_{\mathrm{target}} = \sigma^2_{\overline{m},\mathrm{int}} + [f'(C)\sigma_{C,\mathrm{phot}}]^2 \tag{31}$$

$$+ \sigma^2_{A_{150}} + [f'(C)\sigma_E]^2 - 2f'(C)\sigma_{A_{150}}\sigma_E, \tag{32}$$

where $\sigma_E \equiv \sigma_{E(090-150)}$ and $\mathrm{Cov}(C_0, \overline{m}_{150,0}) = \sigma_{A_{150}}\sigma_E$ because both corrected observables move in the same direction when $E(B-V)$ changes. Its internal predictive variance then separates target measurement, population scatter, and finite calibration:

$$\sigma^2_{\mu,\mathrm{int}} = \sigma^2_{\mathrm{target}} + \sigma^2_{\mathrm{int}} + \sigma^2_{\mathrm{cal}}. \tag{33}$$

The intrinsic term is the value fitted to the relevant 13-object LOO training sample. The finite-calibration term is the standard deviation of 300 non-parametric bootstrap predictions from those same 13 galaxies. The absolute distance error adds the common Paper IV TRGB/NGC 4258 zero point of 0.047 mag. A common F150W photometric zero point and a common PSF-normalization shift cancel to first order between an apparent F150W SBF magnitude and an F150W absolute calibration measured with the same pipeline, so they are not added again as independent distance errors. They do affect the quoted absolute $\overline{M}_{150}$ zero point: the 0.047 mag TRGB, 0.0108 mag flux-scale, and 0.0167 mag PSF-stamp terms combine to 0.0510 mag. The common TRGB term does not enter the relative calibration fit.

Explicitly, a one per cent common F150W flux-scale uncertainty becomes $\sigma_{\mathrm{flux}} = (2.5/\ln 10)(0.01) = 0.0108$ mag. The PSF-stamp term is the largest measured 129-versus-257-pixel spectral shift, 0.0167 mag. Therefore the common absolute-magnitude scale is

$$\begin{aligned}\sigma_{\overline{M},\mathrm{common}} &= \sqrt{0.047^2 + 0.0108^2 + 0.0167^2} \\ &= 0.0510 \text{ mag.}\end{aligned} \tag{34}$$

For a same-filter SBF distance, the last two terms shift apparent and calibrated absolute F150W magnitudes together and cancel; only the 0.047-mag anchor is retained. Any modulus uncertainty is converted to a fractional distance uncertainty with

$$\frac{\sigma_D}{D} = \frac{\ln 10}{5}\sigma_\mu, \tag{35}$$

using the linear form for the quoted percentages (the exact asymmetric conversion is used in the machine-readable distance table).

For the primary SBF-minus-TRGB validation residual, the shared extinction term is $\sigma_{E(090-150)}$. The general color-law diagnostic gives $(1+f')\sigma_{E(090-150)}$; this propagates shared reddening on the difference itself instead of counting it independently in both axes. The validation variance is the quadrature sum of target measurement, target color photometry, this shared-reddening response, individual TRGB error without reddening, intrinsic scatter, and finite-calibration error. LOO uses the same calibrator set and is consequently an internal cross-validation, not an external test on unseen anchors.

Table 2 reports the resulting object-to-object terms, not merely their definitions. The largest median calibration-point component is the individual TRGB error; the exceptional 0.0449-mag sky term belongs to NGC 4486. Table 3 then propagates those terms through an actual LOO distance prediction, and Table 4 separately lists coherent scale errors. Parentheses in the two summary tables give the full minimum–maximum range rather than an uncertainty on the median.

Table 2: Galaxy-dependent statistical uncertainties in magnitudes. The range is the minimum–maximum across the 14 calibration points. Intrinsic and finite-calibration terms are prediction-specific and are shown in Fig. 4.

Component	Median	Range
Power-spectrum fit and k-window	0.0131	0.0095–0.0224
STPSF realization scatter	0.0022	0.0012–0.0033
Residual sources P_r	0.0006	0.0000–0.0013
Sky subtraction	0.0071	0.0035–0.0449
Paper IV TRGB (without reddening)	0.0251	0.0238–0.0271
Shared foreground reddening	0.0040	0.0031–0.0054
Total individual calibration point	0.0302	0.0282–0.0540

5 Primary statistical experiment

The primary result uses all 14 galaxies, individual Paper IV thin-slice distances, no manual rejection, and one color-independent absolute F150W SBF zero point. The linear relation is a pre-specified test of whether a population correction is needed; quadratic and broken-line relations are exploratory. NGC 1399 remains in the primary fit; its constant-zero-point LOO residual is -0.157 mag, or 1.81σ with the validation uncertainty, and does not justify post-hoc removal.

6 Results

6.1 Test for color dependence

We first ask whether color is needed before constructing the primary distance estimator. The color-independent fit gives

$$\overline{M}_{150,0} = -3.172, \qquad \sigma_{\mathrm{int}} = 0.084 \text{ mag.} \tag{36}$$

At the median corrected color $C_0 = 0.5675$, the alternative linear fit is

$$\overline{M}_{150,0} = -3.1756 + 0.4985(C_{090-150,0} - 0.5675), \tag{37}$$

with intrinsic scatter 0.0809 mag. Bootstrap 16th–84th percentiles are $[-3.1975, -3.1486]$ mag for the intercept and $[-0.0576, 0.9931]$ for the slope, so the slope interval includes zero.

The constant model is better than the linear relation by ΔAICc = 2.58 and also has lower LOO RMS. Color therefore does not improve the distance estimate in the present sample. This conclusion is not caused by the treatment of horizontal errors: the linear LOO RMS is 0.09924 mag with no color errors, with the measured errors, or after adding a 0.01-mag floor; using the old half-ring proxy gives 0.09960 mag. We consequently adopt Eq. 36; all subsequent primary distances and error budgets are color independent.

6.2 Distance recovery and precision

The adopted color-independent relation gives LOO bias -0.0004 mag and RMS 0.0955 mag, corresponding to approximately 4.4 per cent scatter in distance. Its median predicted internal uncertainty is 0.0902 mag (4.2

Table 3: Decomposition of a predicted distance-modulus uncertainty. Entries are median (minimum–maximum) in magnitudes across the 14 LOO targets. The constant law has identically zero color propagation. "Internal" is the quadrature of target measurement, color, extinction, intrinsic scatter, and finite-calibration terms. "Absolute" then adds the common 0.047-mag TRGB scale. The final row additionally contains the individual target TRGB error and is used only when validating SBF against TRGB, not when quoting an SBF distance to a new galaxy.

Component	Constant	Linear
Apparent SBF measurement	0.0168 (0.0123–0.0482)	0.0168 (0.0123–0.0482)
Color propagated through calibration	0.0000 (0.0000–0.0000)	1.8×10^{-5} (5.1×10^{-6}–3.2×10^{-5})
Foreground-extinction propagation	0.0030 (0.0023–0.0040)	0.0010 (0.0001–0.0021)
Fitted intrinsic population scatter	0.0848 (0.0770–0.0873)	0.0811 (0.0730–0.0844)
Finite 13-galaxy LOO calibration	0.0245 (0.0223–0.0277)	0.0363 (0.0243–0.0650)
Total internal SBF distance	0.0902 (0.0824–0.1024)	0.0908 (0.0795–0.1160)
Total including common TRGB scale	0.1018 (0.0948–0.1127)	0.1022 (0.0924–0.1252)
SBF–TRGB validation residual	0.0937 (0.0868–0.1053)	0.0946 (0.0843–0.1187)

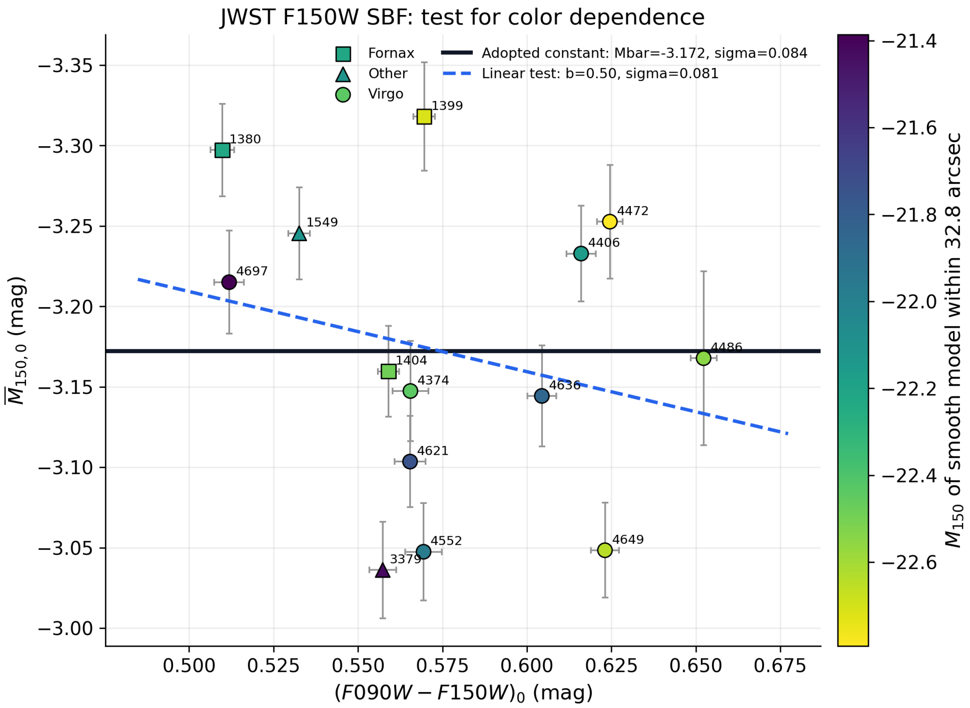

Figure 1: Extinction-corrected JWST F150W SBF calibration and the tested color dependence. The constant zero point is adopted; sloped and subset fits are diagnostics and do not define the primary distances. Marker shape gives environment and marker color the F150W aperture-luminosity proxy.

Table 4: Common systematic uncertainties. The combined absolute $\overline{M}_{150}$ row is the quadrature of the preceding three calibration terms; the same-filter distance row is not added to it.

Component	σ (mag)	Use
TRGB/NGC 4258 zero point	0.0470	absolute calibration
NIRCam F150W flux scale	0.0108	absolute calibration
STPSF stamp normalization	0.0167	absolute calibration
Combined absolute $\overline{M}_{150}$ scale	0.0510	absolute calibration
Same-filter SBF distance scale	0.0470	F150W SBF distances

Table 5: Model comparison for all 14 galaxies. The linear relation tests a possible population correction; nonlinear models are exploratory.

Model	a	b	$\sigma_{\rm int}$	AICc / LOO RMS
Constant	-3.172	0.000	0.084	-22.83 / 0.0955
Linear	-3.176	0.498	0.081	-20.26 / 0.0992
Quadratic	-3.147	0.838	0.074	-18.28 / 0.0989
Broken line	-3.133	2.178	0.073	-18.65 / 0.0974

per cent), and the median absolute uncertainty after the common zero point is 0.1018 mag (4.8 per cent).

The constant-zero-point validation pulls have RMS 1.08; nine of 14 objects lie within 1σ and all 14 within 2σ. Thus the error model is not obviously underestimated or inflated. The empirical LOO RMS and the median predicted uncertainty are distinct quantities and agree at the expected level for this small sample.

Table 6 gives the requested distance comparison in physical units. The constant-zero-point F150W column is the adopted result; the linear column is retained only as the color-dependence test. Both columns are genuine LOO predictions: the listed galaxy was absent when its calibration was fitted. The Paper III F110W values are reconstructed for the eight overlapping objects from its Table 1 and Eq. (2), $\overline{M}_{110} = 1.86[(g-z)_{\rm PS} - 1.30] - 2.760$. Their formal errors propagate the tabulated $\overline{m}_{110}$ and color errors and the published slope and intercept uncertainties as independent terms; common absolute-scale errors are excluded. The last column is the separate Jensen et al. (2015) F160W legacy comparison and has only five overlaps.

6.3 Error distribution by galaxy

Individual calibration-point errors are typically dominated by the TRGB anchor and power-spectrum measurement. NGC 4486 is exceptional because its background term is 0.045 mag. In the predictive distance budget, intrinsic scatter is the largest component, followed by finite-calibration uncertainty; improving the formal P_0 error alone cannot reduce the total distance error below the present 4–5 per cent floor.

6.4 Measurement-pipeline systematics

The local STPSF library contains five unit-normalized models per galaxy. Four offsets around the nominal detector position change the inferred spectral normalization by at most 0.0043 mag. Stamp size matters more: relative to the 129-pixel model, truncation to 97, 65, and 33 pixels can shift the fitted amplitude by as much as 0.0127, 0.0433, and 0.1248 mag, respectively. More importantly, independently generated 257-pixel STPSFs show that the central 129 pixels miss 0.69–0.77 per cent of the 257-pixel flux and imply a common spectral shift of 0.0151–0.0167 mag over the fitted k range. We retain 0.0167 mag as a common absolute-$\overline{M}_{150}$ systematic. Including it with the TRGB and NIRCam terms gives a 0.0510-mag absolute zero-point uncertainty; it does not change the internal LOO residuals.

An empirical check was attempted with six bright, isolated compact sources in the NGC 3379 i2d frame. Their median stack is broader than STPSF and gives a formal 0.288-mag spectral difference. This number is *not* adopted as a PSF correction: the objects are not confirmed single stars, the i2d product has no DQ extension, and residual host/background structure broadens the stack. Thus the model-internal normalization is constrained, but an absolute empirical PSF zero point remains open until an isolated classified star is available. The detailed PSF diagnostics are shown in Appendix A; they are numerical validation plots, not additional astrophysical measurements.

The unresolved-source correction is negligible at the present precision. Across all galaxies, replacing the adopted P_r by either zero or $2P_r$ changes an individual weighted SBF magnitude by at most 0.0053 mag. The constant-model LOO RMS ranges only from 0.0950 to 0.0961 mag, and the linear value from 0.0985 to 0.1001 mag. Re-fitting NGC 1399 and NGC 4472 with the Jensen-specific GCLF width of 1.35 mag changes any weighted magnitude by at most 0.00046 mag. Artificial-source completeness remains desirable, but it cannot alter the present calibration at a scientifically relevant level unless the current P_r estimate is wrong by far more than a factor of two.

A synthetic FFT-stage test passes a known normalized SBF field, mask, PSF, and white-noise term through the FFT and $P_0E(k) + P_1$ fit. Over 32 realizations the recovered magnitude has mean bias 0.0078 ± 0.0067 mag (standard error of the mean), consistent with zero; the trial-to-trial RMS is 0.0382 mag. The test validates the core unit conversion and spectral normalization, but not isophotal modelling, source detection, or the correlated i2d noise. The P_r and FFT-stage diagnostic plots are also collected in Appendix A.

6.5 Distance-anchor sensitivity

Table 7 compares the pre-specified thin-slice analysis, the Paper IV preferred thick-slice values, and a Paper-III-style experiment. The latter assigns the published Paper III means $\mu_{\rm Fornax} = 31.424 \pm 0.057$ mag and $\mu_{\rm Virgo} = 31.055 \pm 0.086$ mag to cluster members; NGC 1549, NGC 3379, and foreground NGC 4697 retain individual anchors.

For the adopted constant calibration, the thin/thick choice changes the zero point by only 0.006 mag and the LOO RMS by less than 0.001 mag. Cluster means reduce the fitted intrinsic scatter, as expected when depth and individual TRGB noise are suppressed, yet the gain against individual thin-slice distances is small.

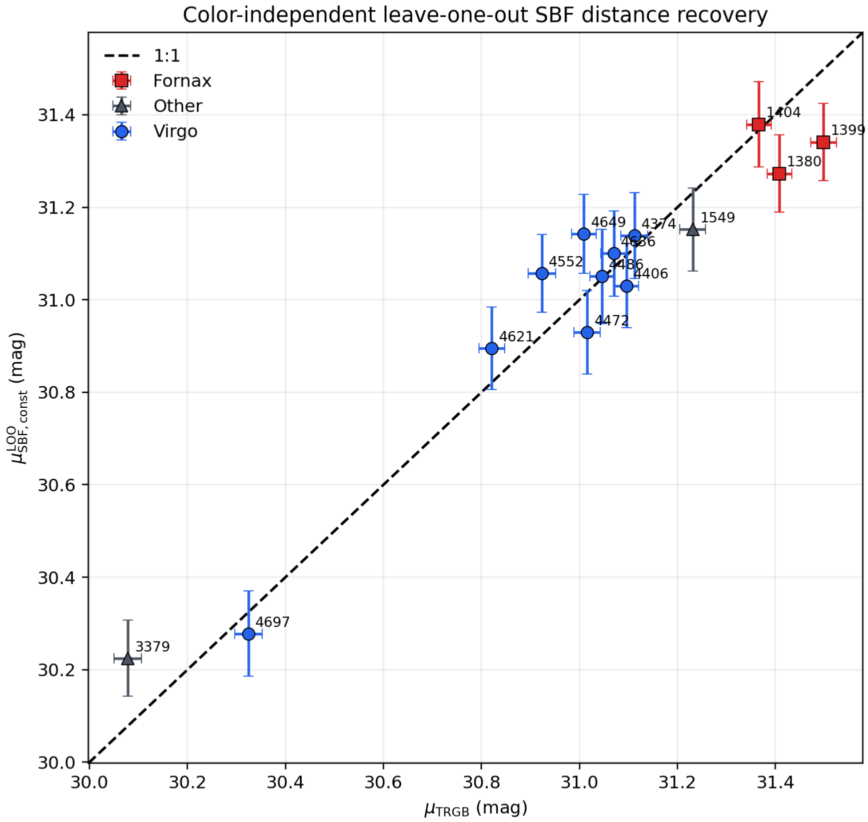

Figure 2: Color-independent LOO F150W SBF distance moduli compared with the individual Paper IV thin-slice TRGB moduli. Each target is excluded from its own calibration. Error bars are internal; the common absolute zero point is not shown.

Table 6: Distances in Mpc. TRGB IV denotes the individual Paper IV thin-slice anchor. The primary JWST F150W SBF column uses the constant LOO calibration; the linear LOO column is diagnostic. Jensen III is the Paper III HST F110W reconstruction; Jensen 2015 is the independently labelled HST F160W legacy result. Ellipses mean that the galaxy is absent from that comparison sample. Quoted errors exclude common absolute zero points.

Galaxy	Env.	TRGB IV	SBF const.	SBF linear	Jensen III F110W	Jensen 2015 F160W
NGC 1380	Fornax	19.12 ± 0.22	17.97 ± 0.69	18.05 ± 0.79	18.98 ± 0.31	20.12 ± 1.19
NGC 1399	Fornax	19.93 ± 0.24	18.54 ± 0.71	18.56 ± 0.69	19.80 ± 0.34	20.30 ± 1.19
NGC 1404	Fornax	18.76 ± 0.22	18.88 ± 0.80	18.95 ± 0.79	19.44 ± 0.33	19.84 ± 1.22
NGC 1549	Other	17.63 ± 0.21	17.00 ± 0.70	17.13 ± 0.75	...	...
NGC 3379	Other	10.37 ± 0.13	11.09 ± 0.42	11.15 ± 0.41	...	...
NGC 4374	Virgo	16.69 ± 0.21	16.89 ± 0.72	16.93 ± 0.70	...	...
NGC 4406	Virgo	16.57 ± 0.19	16.07 ± 0.67	15.85 ± 0.66	...	...
NGC 4472	Virgo	15.97 ± 0.20	15.34 ± 0.64	15.05 ± 0.63	16.11 ± 0.27	16.19 ± 0.98
NGC 4486	Virgo	16.19 ± 0.19	16.22 ± 0.77	15.84 ± 0.85	...	...
NGC 4552	Virgo	15.30 ± 0.20	16.27 ± 0.63	16.29 ± 0.61	15.70 ± 0.27	...
NGC 4621	Virgo	14.59 ± 0.17	15.10 ± 0.62	15.14 ± 0.61	...	...
NGC 4636	Virgo	16.37 ± 0.20	16.60 ± 0.71	16.48 ± 0.71	16.30 ± 0.27	...
NGC 4649	Virgo	15.91 ± 0.18	16.92 ± 0.66	16.83 ± 0.72	16.19 ± 0.31	15.86 ± 0.98
NGC 4697	Virgo	11.61 ± 0.15	11.36 ± 0.48	11.53 ± 0.56	11.66 ± 0.19	...

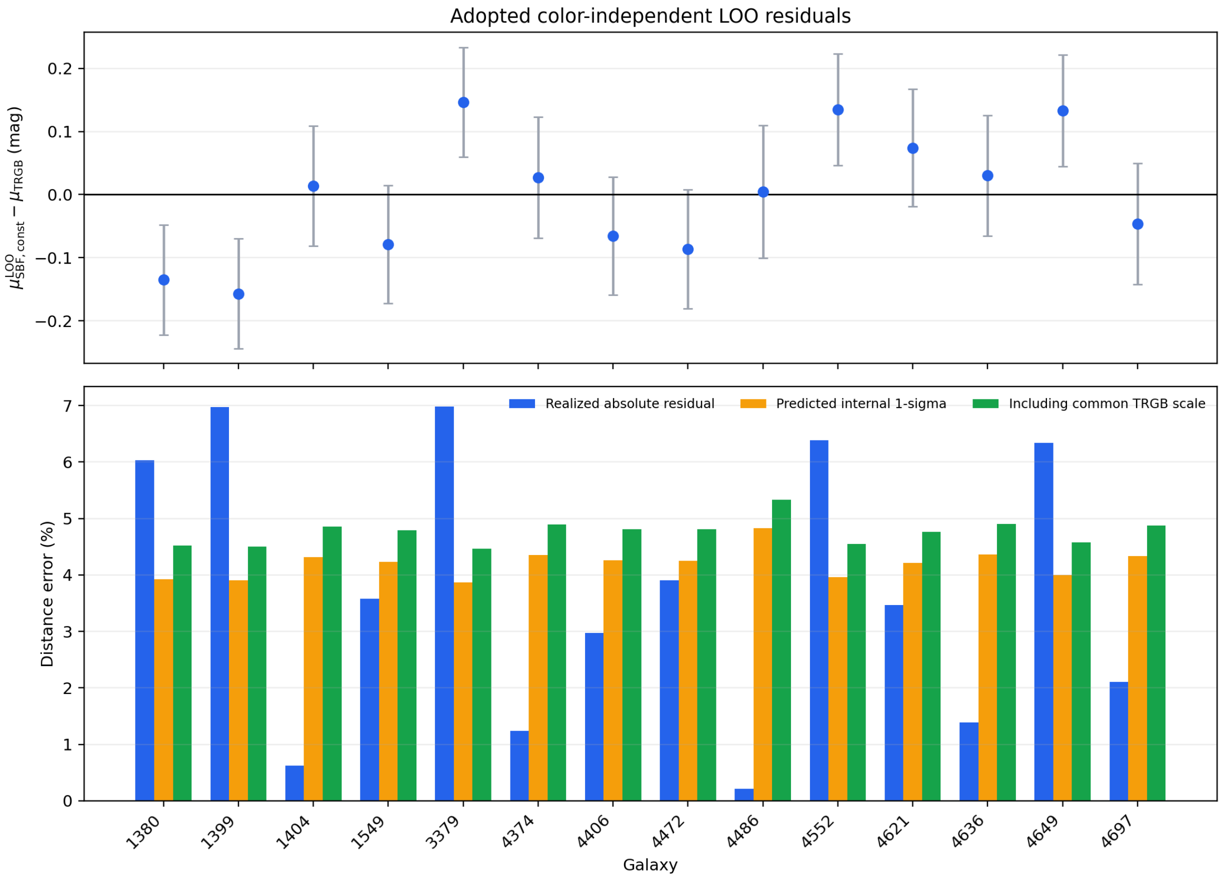

Figure 3: Top: color-independent LOO SBF-minus-TRGB residuals with the propagated validation uncertainty. Bottom: realized absolute distance residual, predicted internal uncertainty, and predicted uncertainty after the common absolute zero point, for every galaxy.

Table 7: Sensitivity to the TRGB distance prescription. The final column always evaluates LOO predictions against the primary individual Paper IV thin-slice distances. Cluster-mean LOO values are not an independent galaxy-by-galaxy validation because different galaxies share an anchor.

Anchor	Model	a	b	$\sigma_{\rm int}$	LOO RMS vs anchor	LOO RMS vs thin
Paper IV thin	constant	-3.172	0.000	0.084	0.0955	0.0955
Paper IV thin	linear	-3.176	0.498	0.081	0.0992	0.0992
Paper IV thick	constant	-3.166	0.000	0.083	0.0953	0.0956
Paper IV thick	linear	-3.170	0.597	0.079	0.0988	0.1000
Paper III cluster means	constant	-3.196	0.000	0.068	0.0931	0.0980
Paper III cluster means	linear	-3.194	0.570	0.065	0.0937	0.0970

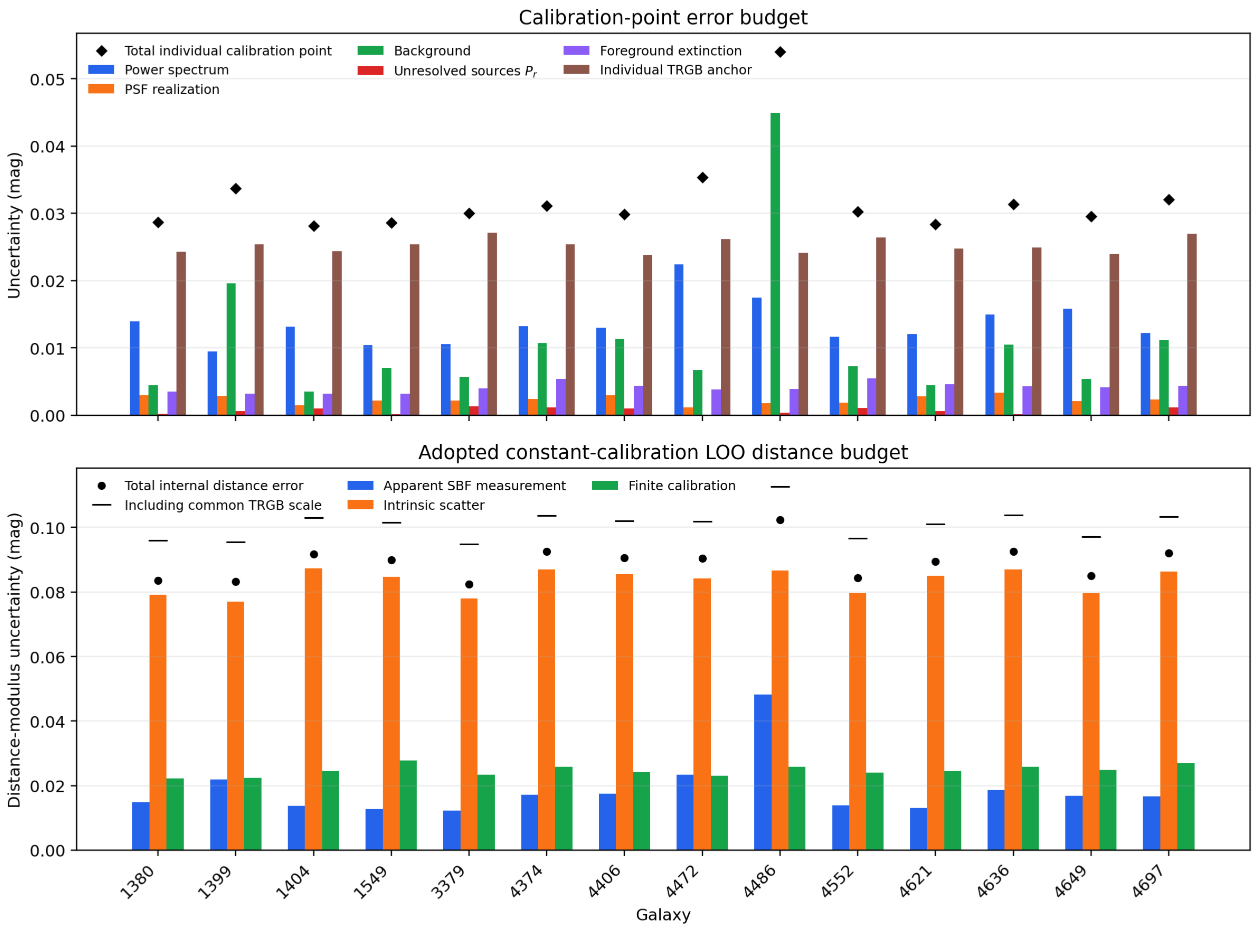

Figure 4: Top: individual calibration-point error components. The PSF bar is propagated separately from the power-spectrum measurement. Bottom: color-independent LOO distance-error components. Black circles are total internal errors and horizontal ticks include the common absolute zero point.

No conclusion depends on selecting the thin variant over the Paper IV preferred thick estimator. Linear rows are retained in the table only to show that the rejected color correction does not become necessary under another anchor prescription.

6.6 Comparison with Paper III colors

For our 14 galaxies, the correlation of $\overline{M}_{150,0}$ with $(F090W - F150W)_0$ is weak. The 14 observed Paper III F110W calibration points show a much stronger correlation with $g - z$ over their wider color range. The two colors themselves agree well for the eight common galaxies: Pearson $r = 0.89$ ($p = 0.003$) and Spearman $\rho = 0.88$ ($p = 0.004$). Thus the weak F150W slope cannot be attributed to a complete failure of the infrared color as a population tracer; the sample and wavelength response differ.

6.7 Radial behavior and environment

The inner annuli give slope 1.24 and intrinsic scatter 0.068 mag; the outer annuli give slope 0.08 and scatter 0.089 mag. The two points per galaxy are correlated and are not 28 independent calibrators. NGC 3379 has an exceptionally uniform residual image but a 0.168 mag annular SBF difference, demonstrating why the annular difference cannot be treated automatically as random measurement noise.

For the strict Virgo/Fornax subset (7+3 galaxies), a fitted Fornax shift is -0.150 mag. However, an exact permutation test gives $p = 0.183$, color and cluster membership are strongly confounded, and the step model is worse by $\Delta\text{AICc} = 1.76$. The step is a diagnostic, not a detection. Foreground extinction does not remove this pattern.

6.8 Structural quality and aperture magnitude

Quality flags were declared independently of the calibration fit. NGC 4486 contains the jet and strong large-scale structure; NGC 4472, NGC 4636, and NGC 4649 have strong residual structure; and NGC 1380 and NGC 4697 have incomplete outer model coverage. The primary sample nevertheless remains 14/14. Restricting it to the 11 visually acceptable or 10 automatically clean objects does not improve cross-validation: the constant-model LOO RMS becomes 0.0983 and 0.1034 mag, respectively, compared with 0.0955 mag for all 14. The test supports retaining the full sample while reporting the flags rather than using image appearance for post-hoc rejection.

The plotted host quantity is the absolute F150W magnitude of the smooth model within 32.8 arcsec, not a total galaxy luminosity. The fixed angular aperture subtends different physical radii and therefore can generate an apparent Virgo–Fornax offset. No direct correlation is detected for all 14 objects (Pearson $p = 0.220$), for Virgo alone ($p = 0.910$), or for the three Fornax objects ($p = 0.954$). After subtracting the rejected linear color relation as a diagnostic, the corresponding p values are 0.083, 0.514, and 0.836. The residual panel is therefore an exploratory partial-correlation diagnostic, not part of the adopted distance estimator.

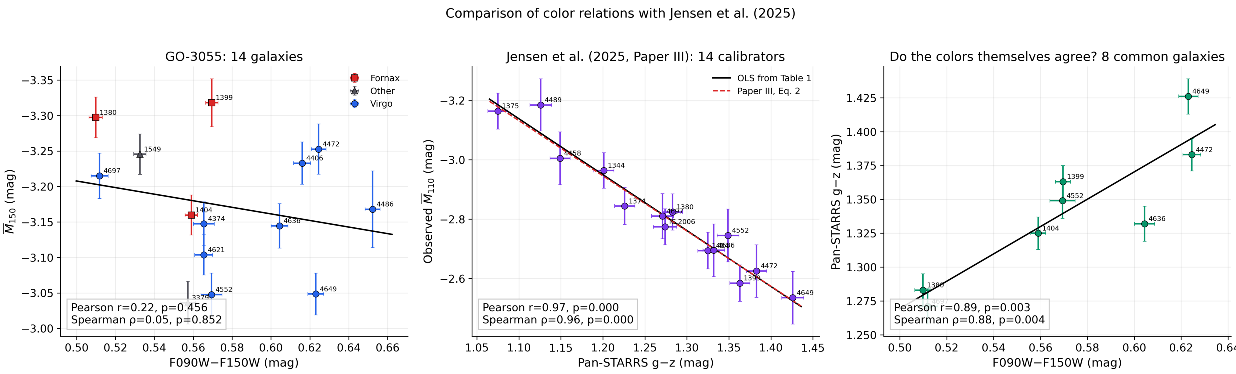

Figure 5: SBF–color relations in GO-3055 and Paper III, followed by the direct color-to-color comparison for eight common galaxies. Slopes are not numerically interchangeable because both the SBF band and color index differ.

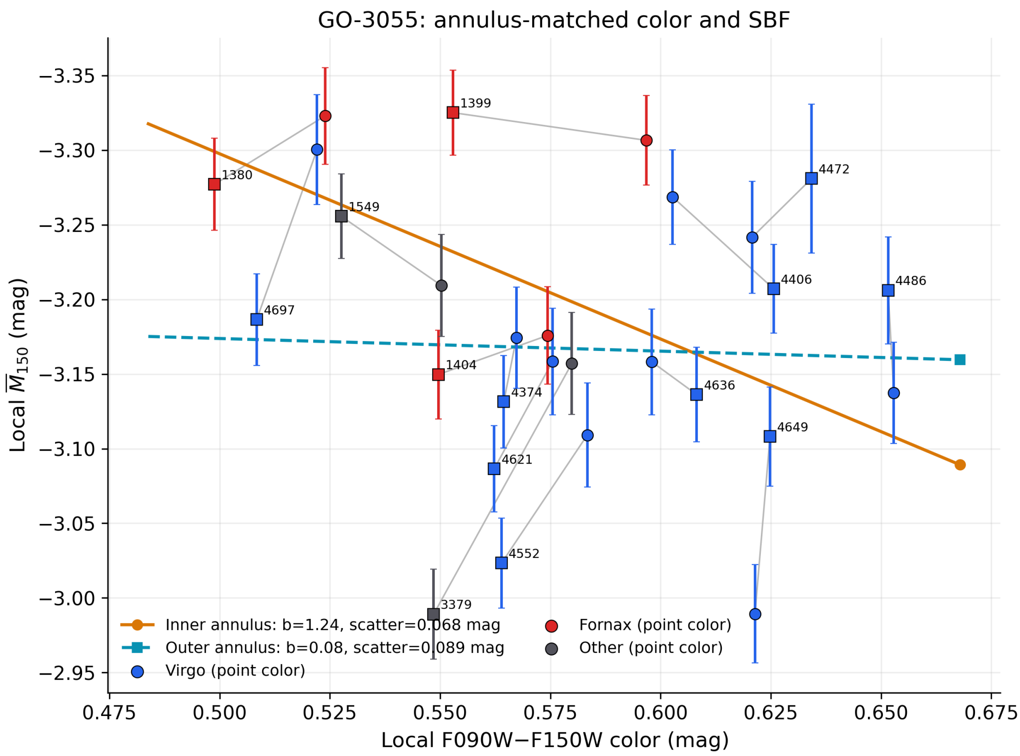

Figure 6: Local color and absolute F150W SBF magnitude in matched annuli. Grey segments connect the two measurements of each galaxy.

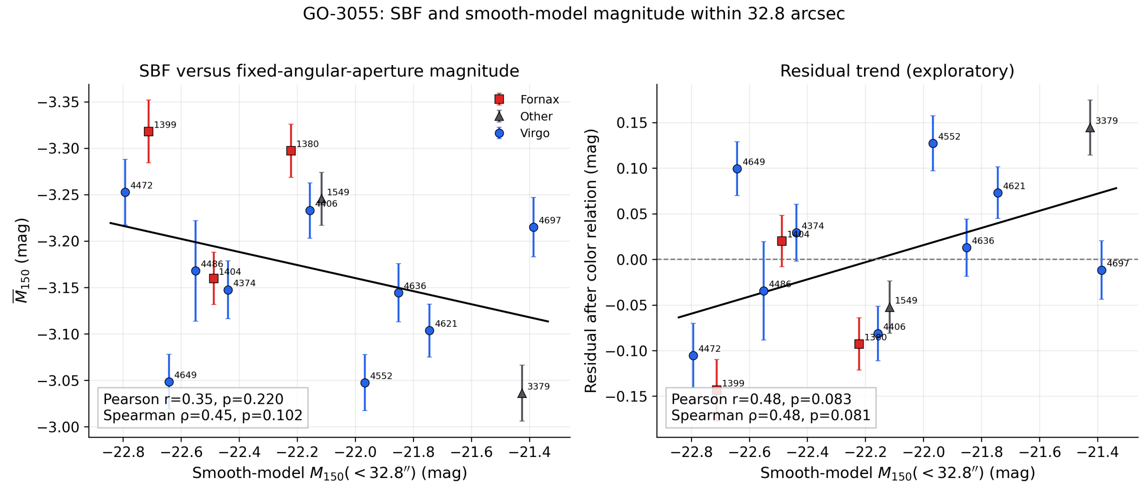

Figure 7: F150W SBF versus the smooth-model magnitude inside the fixed 32.8-arcsec aperture. The right panel subtracts the rejected linear color relation as a diagnostic; it does not subtract the regression drawn in the left panel and is not used for distances. Neither trend is statistically detected.

7 Discussion

7.1 Accuracy of the method

The central empirical result is a cross-validated F150W SBF precision of approximately 4.4 per cent in distance for this 14-galaxy sample using one color-independent F150W zero point. The tested color correction does not improve this result and is not used in the adopted distances.

The uncertainty model passes a useful internal calibration check: the LOO pull RMS is close to unity and no target exceeds 2σ. This does not prove the absence of shared pipeline systematics. LOO galaxies reuse the same instrument, reduction, PSF construction, and TRGB scale; only a separate galaxy sample can provide an external accuracy test.

7.2 Remaining limitations

Three measurement systematics remain physically important. First, i2d noise is correlated and has not yet been fitted with an explicit $N(k)$ template. Second, the retained 3.5σ winsorization changes a median 3.35 per cent of inner-ring pixels and shifts a full-variance proxy by -0.092 mag. That proxy is not the fitted P_0; exact refits without winsorization and at several limits are required before treating the operation as negligible. Third, the empirical PSF comparison is not closed by a confirmed isolated star. The compact-source completeness and P_r model also remain approximate; however, the explicit 0–$2P_r$ test bounds their impact on this calibration, so artificial-source injection can be deferred without claiming that completeness itself has been measured.

8 Conclusions

1. All 14 GO-3055 galaxies are retained in the primary analysis with individual Paper IV thin-slice TRGB anchors and no manual exclusions.
2. A constant F150W absolute SBF magnitude recovers the TRGB distances with 0.0955 mag LOO RMS (4.4 per cent) and negligible mean bias.
3. The median predicted color-independent precision is 4.2 per cent internally and 4.8 per cent after the common TRGB zero point. The observed pull distribution is consistent with the adopted errors.
4. The data do not require a color dependence: a linear color law worsens the LOO RMS to 0.0992 mag, its AICc is worse by 2.58, and its slope interval includes zero. It is therefore retained only as a diagnostic and is not used for the primary distances.
5. Paper IV thick-slice distances and Paper III cluster means do not materially change the recovered accuracy. Thin slices remain primary only because that experiment was pre-specified, not because they give a better answer.
6. The Virgo–Fornax step and NGC 1399 exclusion remain diagnostics; neither is promoted to a detected effect or used to define the main sample.
7. The P_r sensitivity and synthetic FFT tests are negligible at the current distance precision. The 129-pixel PSF stamp instead introduces a 0.015–0.017 mag common absolute-calibration term.
8. The result is an internal LOO cross-validation. Correlated noise, winsorization, and a confirmed empirical stellar PSF are still required before claiming final external accuracy.

A Measurement-pipeline diagnostic plots

The following plots document numerical sensitivity tests requested during pipeline validation. They support the systematic-error bounds quoted in the main text and do not enter the adopted color-independent distance estimator as additional data.

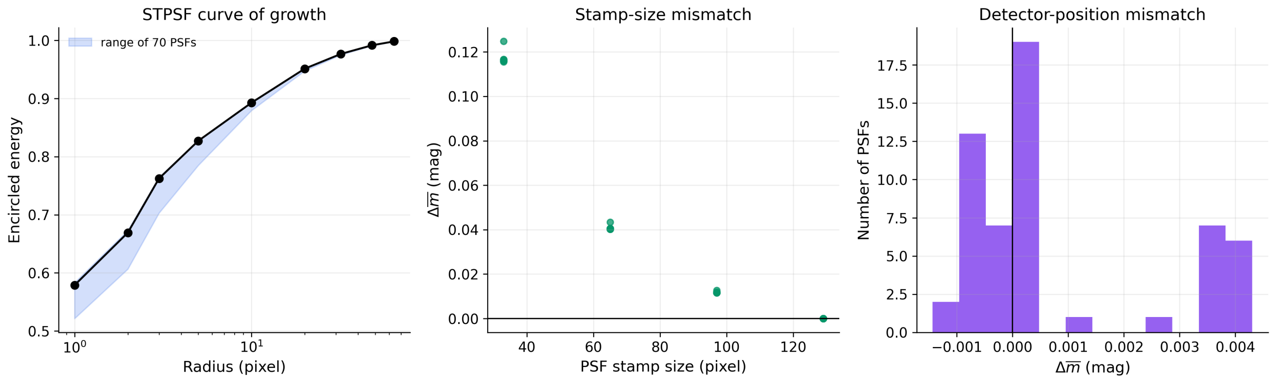

Figure 8: Internal PSF tests. Left: STPSF encircled-energy curves for all 70 models. Middle: spectral shifts caused by truncating the saved 129-pixel model. Right: shifts from the four detector-position variants.

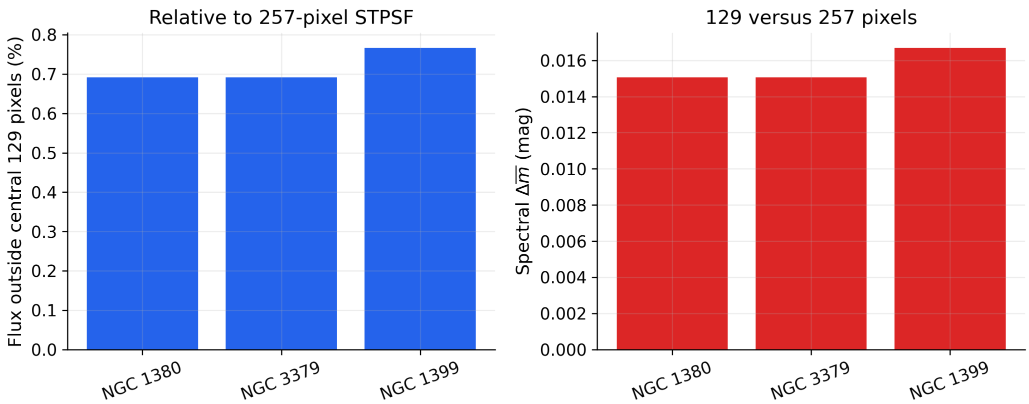

Figure 9: Absolute stamp-size test for three representative targets. The left panel gives flux outside the central 129 pixels relative to a 257-pixel STPSF; the right panel translates the shape mismatch over the fitted frequency range into an SBF-magnitude shift.

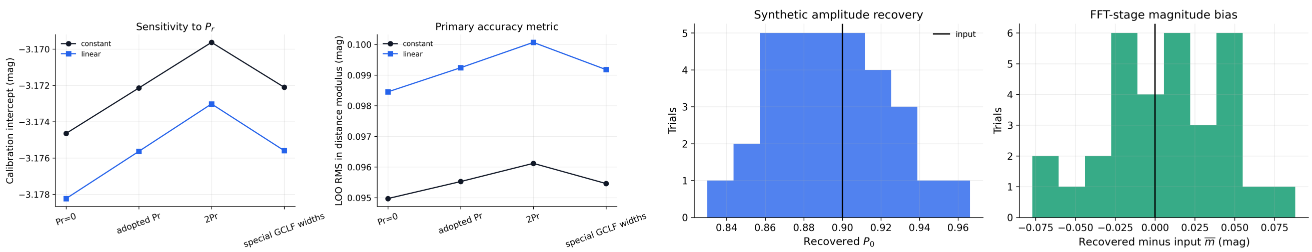

Figure 10: Left: calibration and color-independent LOO metric under the P_r sensitivity experiments. Right: recovery of known P_0 and SBF magnitude in the synthetic FFT-stage test.

B Machine-readable products

The full distance table is `runs/sbf2_go3055/analysis/tables/go3055_distances_mpc.csv`. The joined TRGB/F150W/F110W/F160W comparison is `runs/sbf2_go3055/analysis/tables/go3055_distance_comparison_all_methods.csv`. The predictive errors are in `runs/sbf2_go3055/analysis/tables/go3055_distance_predictions_full_errors.csv`; the component table is `runs/sbf2_go3055/analysis/tables/go3055_error_budget.csv`; anchor tests are in `runs/sbf2_go3055/analysis/tables/go3055_distance_anchor_sensitivity.csv`.

Additional machine-readable systematics products are `runs/sbf2_go3055/analysis/tables/go3055_psf_129_vs_257_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_psf_detector_position_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_Pr_calibration_sensitivity.csv`, `runs/sbf2_go3055/analysis/tables/go3055_synthetic_fft_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_structural_quality_and_physical_annuli.csv`. The compact uncertainty summaries printed in the article are also available as `runs/sbf2_go3055/analysis/tables/go3055_individual_error_summary.csv`, `runs/sbf2_go3055/analysis/tables/go3055_predictive_error_summary.csv`, and `runs/sbf2_go3055/analysis/tables/go3055_common_systematics.csv`.

C Data and code availability

The calibrated JWST observations are public through MAST under GO-3055; the archive collection is accessible through DOI 10.17909/z9ch-wk24. No proprietary data are used. The machine-readable derived products required to reproduce the tables and figures are listed above and retained with this working paper.

The measurement and analysis code is presently the version-controlled project copy described below. A public archival release and permanent code DOI have not yet been assigned; they must be added before submission. This statement does not falsely claim that an unarchived local repository is publicly available.

D Reproducibility

The measurement notebook is `code/sbf-2.ipynb`; the dedicated GO-3055 runner is `code/run_sbf_2_batch.py`. Exploratory fits and diagnostic figures are generated and displayed inline by `code/sbf-2-graph.ipynb`, which reads completed batch products and does not repeat the SBF measurement. The adopted constant-calibration tables, distance-recovery figures, error-budget figure, comparison tables, and provenance records are generated from those saved products by `code/build_sbf2_article_tables.py`.

The implementation uses NumPy, SciPy, Astropy, Photutils, Matplotlib, and STPSF [17–23]. Exact installed versions, rather than only literature citations, are given in Table 10.

The 28 input i2d headers are heterogeneous archive products rather than a uniform local reprocessing: 2 files record JWST pipeline 1.19.1, CRDS 12.1.11, and `jwst_1413.pmap`; 22 record pipeline 1.20.2, CRDS 13.0.6, and `jwst_1464.pmap`; and 4 F090W files record pipeline 2.0.1, CRDS 13.1.13, and `jwst_1535.pmap`. Exact counts are generated directly from the files used by the batch and retained in `go3055_input_calibration_provenance.csv`. This records the reduction state exactly, but does not prove that mixing contexts is harmless; uniform reprocessing is a residual reproducibility/systematics check. The product-generation run records Git HEAD `5d0fd1496474...` and the exact executed `sbf-2.ipynb` SHA256 `a1828d2d8d97...`. Full identifiers are retained in `go3055_software_provenance.csv`. Its provenance also records that the worktree was dirty. Consequently the commit alone is not sufficient; the saved notebook snapshot and its hash are the authoritative measurement recipe. The final article commit/tag must be inserted after the present edits are committed.

From the project root, the reproducible analysis commands are

```
export MPLBACKEND=Agg
astro_env/bin/jupyter nbconvert \
  --to notebook --execute --inplace \
  code/sbf-2-graph.ipynb \
  --ExecutePreprocessor.timeout=3600
astro_env/bin/python \
  code/build_sbf2_article_tables.py
```

The Agg backend is mandatory for non-interactive execution on macOS; figures are saved and also displayed inline in the notebook.

Table 8: Predeclared structural-quality diagnostics. The automatic flag is set at a structure score of 0.80; the visual score is on the user-defined 0–10 scale. These flags define sensitivity samples and do not remove any galaxy from the primary 14-object calibration.

Galaxy	Visual score	Auto score	Auto flag	Coverage flag	Predeclared issue
NGC 1380	7.00	0.40	False	True	incomplete outer model coverage
NGC 1399	9.50	0.05	False	False	
NGC 1404	8.00	0.12	False	False	
NGC 1549	9.00	0.18	False	False	
NGC 3379	10.00	0.26	False	False	
NGC 4374	8.00	0.15	False	False	
NGC 4406	8.00	0.64	False	False	
NGC 4472	4.00	1.61	True	False	strong large-scale structure
NGC 4486	3.00	1.47	True	False	jet residual; strong large-scale structure
NGC 4552	9.00	0.20	False	False	
NGC 4621	7.00	0.35	False	False	
NGC 4636	6.00	0.97	True	False	strong large-scale structure
NGC 4649	5.00	1.40	True	False	strong large-scale structure
NGC 4697	6.00	0.40	False	True	incomplete outer model coverage

Table 9: Fixed processing configuration for the reported run. Values not listed here remain recorded in the executed notebook snapshot.

Stage	Adopted configuration
Input and bands	JWST/NIRCam i2d; F150W SBF signal; F090W–F150W color; native pixel scale $\simeq 0.03123$ arcsec
Science regions	circular 8.2–16.4 and 16.4–32.8 arcsec annuli
Isophotes	initial semimajor axis 50 pixels; free ellipticity and position angle; 10-pixel sampling; maximum 2000 pixels; real-pixel fit with filled-gap fallback
Compact masks	primary 3.5σ, 9 connected pixels, 2-pixel dilation; residual catalogue 2.5σ, 4 connected pixels, deblending, 3-pixel dilation
Residual limiting	symmetric 3.5σ winsorization; raw residual retained
PSF	STPSF; nearest time-matched WSS OPD; nominal plus four detector offsets; 129 pixels; 7 wavelengths; unit normalization
Fourier model	80 radial bins; 64 Monte Carlo realizations per $E(k)$ and PSF; $P_0E(k) + P_1$; primary $k = 0.04$–0.25; $k_{\min} = 0.01, 0.03$ sensitivity windows
Unresolved sources	Gaussian GCLF plus background-galaxy power law; 12-mag faint integration; 25 per cent P_r uncertainty; 0, P_r, and $2P_r$ tests
Calibration	all 14 primary; individual Paper IV thin-slice anchors; constant zero point with fitted intrinsic scatter; 300 LOO prediction bootstraps. The linear errors-in-variables fit with reddening covariance and 2000 fit bootstraps is diagnostic only

Table 10: Software and immutable identifiers. Long hashes are abbreviated in print and retained in full in the accompanying CSV.

Item	Version or identifier
Input i2d products (2/28)	pipeline 1.19.1; CRDS 12.1.11; jwst_1413.pmap
Input i2d products (22/28)	pipeline 1.20.2; CRDS 13.0.6; jwst_1464.pmap
Input i2d products (4/28)	pipeline 2.0.1; CRDS 13.1.13; jwst_1535.pmap
Python	3.13.2
numpy	2.3.5
astropy	7.2.0
photutils	2.3.0
stpsf	2.1.0
scipy	1.16.3
matplotlib	3.10.8
pandas	2.3.3
Product-generation Git HEAD	`5d0fd1496474...`
Executed sbf-2.ipynb SHA256	`a1828d2d8d97...`
Current sbf-2-graph.ipynb SHA256	`5953cc91e304...`
Current article TeX SHA256	`6e49f322f50b...`
Article-table builder SHA256	`16c6b21432f9...`
Current article-worktree Git HEAD	`3ea8f14fcc58...`
Current article worktree clean	no

References

[1] Tonry, J. L., & Schneider, D. P. 1988, AJ, 96, 807.

[2] Tonry, J. L., Dressler, A., Blakeslee, J. P., et al. 2001, ApJ, 546, 681, doi:10.1086/318301.

[3] Blakeslee, J. P., Lucey, J. R., Tonry, J. L., et al. 2002, MNRAS, 330, 443, doi:10.1046/j.1365-8711.2002.05080.x.

[4] Blakeslee, J. P., Jordan, A., Mei, S., et al. 2009, ApJ, 694, 556, doi:10.1088/0004-637X/694/1/556.

[5] Blakeslee, J. P., Cantiello, M., Mei, S., et al. 2010, ApJ, 724, 657, doi:10.1088/0004-637X/724/2/657.

[6] Blakeslee, J. P., Jensen, J. B., Ma, C.-P., Milne, P. A., & Greene, J. E. 2021, ApJ, 911, 65, doi:10.3847/1538-4357/abe86a.

[7] Cantiello, M., Jensen, J. B., Blakeslee, J. P., et al. 2018, ApJL, 854, L31, doi:10.3847/2041-8213/aaad64.

[8] Cantiello, M., Blakeslee, J. P., Ferrarese, L., et al. 2024, ApJ, 966, 145, doi:10.3847/1538-4357/ad3453.

[9] Jensen, J. B., Tonry, J. L., Thompson, R. I., et al. 2001, ApJ, 550, 503, doi:10.1086/319819.

[10] Jensen, J. B., Tonry, J. L., Barris, B. J., et al. 2003, ApJ, 583, 712, doi:10.1086/345430.

[11] Jensen, J. B., Blakeslee, J. P., Ma, C.-P., et al. 2015, ApJ, 808, 91, arXiv:1505.00400.

[12] Jensen, J. B., Blakeslee, J. P., Ma, C.-P., et al. 2021, ApJS, 255, 21.

[13] Jensen, J. B., Blakeslee, J. P., Cantiello, M., et al. 2025, *The TRGB–SBF Project. III. Refining the HST Surface Brightness Fluctuation Distance Scale Calibration with JWST*, ApJ, arXiv:2502.15935v2.

[14] Chazov, M. I., Makarov, D. I., Tully, R. B., et al. 2026, *The TRGB–SBF Project. IV. A Color Calibration of the TRGB in the JWST F090W+F150W Filters*, arXiv:2603.11160v1.

[15] Schlafly, E. F., & Finkbeiner, D. P. 2011, ApJ, 737, 103.

[16] Reid, M. J., Pesce, D. W., & Riess, A. G. 2019, ApJL, 886, L27, doi:10.3847/2041-8213/ab552d.

[17] Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33, doi:10.1051/0004-6361/201322068.

[18] Astropy Collaboration, Price-Whelan, A. M., Sipocz, B. M., et al. 2018, AJ, 156, 123, doi:10.3847/1538-3881/aabc4f.

[19] Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi:10.3847/1538-4357/ac7c74.

[20] Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi:10.1038/s41586-020-2649-2.

[21] Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261, doi:10.1038/s41592-019-0686-2.

[22] Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi:10.1109/MCSE.2007.55.

[23] Perrin, M. D., Sivaramakrishnan, A., Lajoie, C.-P., et al. 2014, Proc. SPIE, 9143, 91433X, doi:10.1117/12.2056689.